Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026

Instructor – Dr. Aamir Hasan - Section L4

Time = 20 Minutes **Quiz No 06** [20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning.

CLO Assessed – CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3)

Question 1 [20 points] [Cumulative Distribution Function for Normal Random Variables]

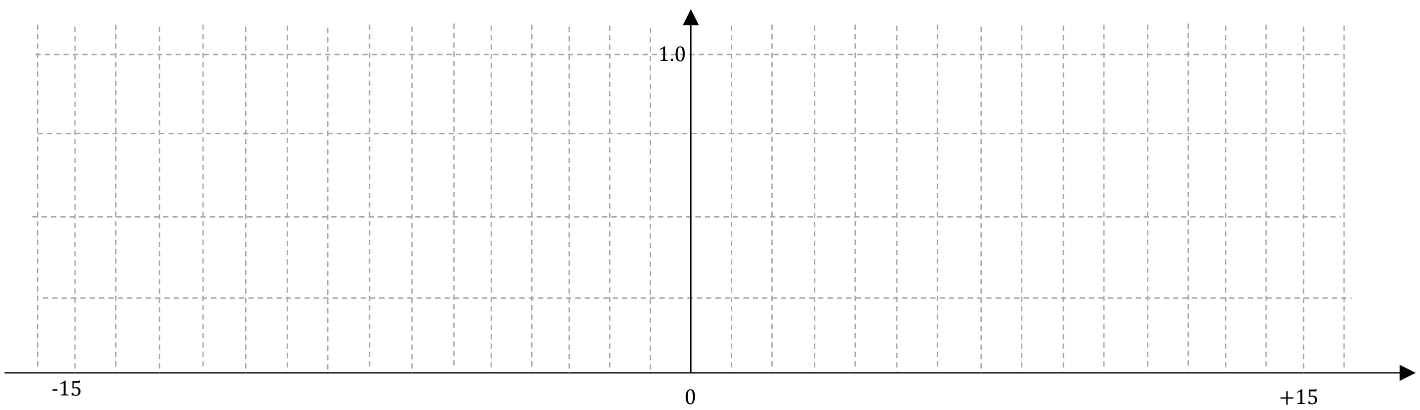
We have the following four Normal Random Variables, $r.v \sim \boldsymbol{N}(mean, variance)$ that are independent to each other:

- $X \sim \boldsymbol{N}(0,1)$
- $Y \sim \boldsymbol{N}(0,4)$
- $W \sim \boldsymbol{N}(4,1)$
- $Z \sim \boldsymbol{N}(-4,9)$

Using the given one plot (*even at the cost of slight abuse of notations*) below, plot the Cumulative Distribution Functions for the four Normal R.Vs given above. *(You may use the normal standard table provided on the next page).*

Note – while exact CDF values are not required for the entire range but the ones that are critical (to show your understanding and correctness) must be captured. Also, the trends, notations and comparisons need to be made for the 4 CDFs. Include enough details, labelling and arguments. Also show critical & needed values on the x-axis and y-axis on the single plot wisely. ***Grading - 12 points are for the 4 plots and 8 points for the explanation.***

Hint – if you have a good grip on the CDF for r.v. $\boldsymbol{X}$, then the remaining 3 CDFs can be drawn accordingly (in reference to) showing how the other 3 normal random variables CDF differ to that of $\boldsymbol{X}$ and also, amongst each other.



Spare plot – in case you mess the earlier plot